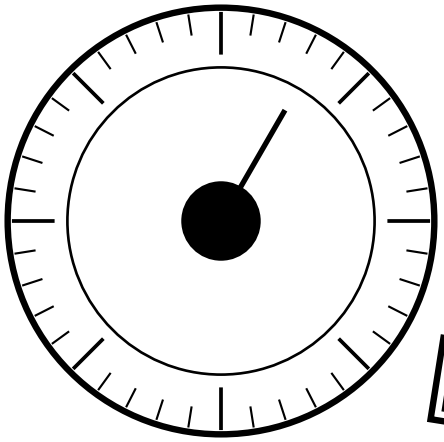
BONUS MISSION PACK

Ten extra missions for agents who finished the book,
or who just want more. Missions 81 to 90.
Hints, check codes and full solutions are inside.



FOR AGENTS ONLY

FUN BOOKS
PUBLISHER

LOGIC GRID | LEVEL ■■□ ADVANCED TYPICAL TIME: 14 MIN

SECTOR ATLAS SORT-OUT

Headquarters lost the roster file during last night's system update. These notes are the only backup copy you have left.

RULES Each agent matches exactly one item in every category, and no two agents share an item. Mark a box with X for no and O for yes. Earlier, later, lower and higher mean by any amount unless the clue says exactly.

		ROBOT				CODE NAME				FLOOR			
		Bolt	Gear	Rivet	Spark	Comet	Otter	Lynx	Orca	Floor 1	Floor 2	Floor 3	Floor 4
AGENT	Isaac												
	Maya												
	Sadie												
	Kiara												
FLOOR	Floor 1												
	Floor 2												
	Floor 3												
	Floor 4												
CODE NAME	Comet												
	Otter												
	Lynx												
	Orca												

1. Sadie works with the robot Spark.
2. The agent on Floor 1 is either Sadie or Maya.
3. The agent with the robot Gear works on a lower floor than Kiara.
4. The agent with the robot Gear is either Sadie or Isaac.
5. Isaac is not code-named Lynx.
6. The agent with the robot Bolt works on Floor 2.
7. The agent code-named Orca works on Floor 1.
8. The agent on Floor 2 is code-named Otter.

Which agent is code-named Lynx?

ANSWER _____ CHECK CODE (PAGE 12) _____

KEYWORD CIPHER | LEVEL ■■■ ELITE TYPICAL TIME: 21 MIN

CODE NAME: BROKEN BEACON

Your trainer, Agent Marcus, wrote this test message. Prove you can read it without any help.

RULES | The secret alphabet starts with the keyword, with repeated letters left out, then the rest of A to Z in order. Each plain letter sits above its secret letter.

INTERCEPTED MESSAGE

V	W	P	L	C		P	C	U		O	F	Y

W	A	J	T	U		L	W	A	D	I

U	D	B	C	P

DECODER KEY

Keyword: WALRUS

PLAIN	A	B	C	D	E	F	G	H	I	J	K	L	M
SECRET													

PLAIN	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
SECRET													

What is the first word of the message?

ANSWER | CHECK CODE (PAGE 12)

LOCK CODE | LEVEL ■■□ ADVANCED TYPICAL TIME: 10 MIN

THE RUSTY TUMBLER

The new air-quality sensor sits behind a dial with no instructions. Layla kept notes on every failed try.

RULES | Each clue is an earlier try at the code. Its sentence says how many of its digits are in the code and how many of those are in the right spot. Any other digit of that try is not in the code. The code never repeats a digit.

	TRY	WHAT THE LOCK SAYS
1	7 2 5	One digit is correct and in the right spot.
2	1 4 7	One digit is correct but in the wrong spot.
3	9 6 5	Two digits are correct: one in the right spot, one in the wrong spot.
4	0 9 2	Nothing is correct.



SPOT TRACKER

Cross out a digit at the top once it is out.
Put an X where a digit cannot go, a circle where it must.

DIGIT	0	1	2	3	4	5	6	7	8	9
1ST										
2ND										
3RD										

What is the 3-digit code?

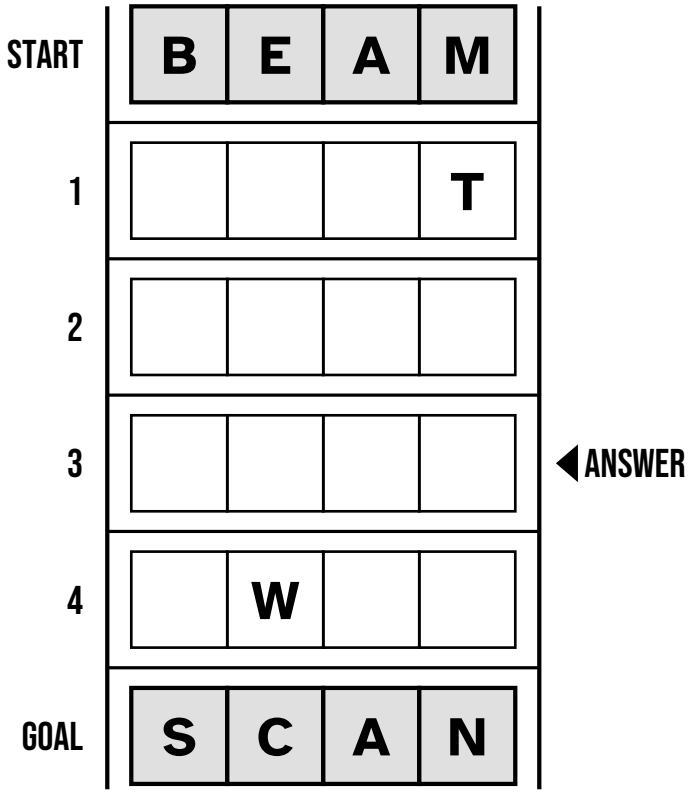
ANSWER _____ CHECK CODE (PAGE 12) _____

WORD LADDER | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

SIGNAL PATH: BEAM TO SCAN

A message drone dropped half its cargo. Use the printed letters to rebuild the climb from BEAM to SCAN.

RULES | Change exactly one letter at each step to climb from the top word to the bottom word. Every rung must be a real English word, and no word can appear twice. Printed letters are already in place.



WORK SPACE

What word sits on rung 3 of the ladder?

ANSWER _____ CHECK CODE (PAGE 12) _____

CRYPTOGRAM | LEVEL ■■□ ADVANCED TYPICAL TIME: 15 MIN

RADIO LOG: COBALT TIDE

The ocean lab sends its notes in code so no one can copy the results. Decode the newest note.

RULES | Each secret letter always stands for the same real letter, and no letter stands for itself. The shaded boxes give you three letters to start. Use the tracker to keep track.

INTERCEPTED MESSAGE

Shaded boxes are given.

G	A	H		K	V	W	D	U		M	D	R	H	W

C	V	U	B	K		D	Q	D	O	R		G	H	K	G
			K				G				.				

G	A	H		R	H	C		J	D	G	G	H	U	I
								B						

G	V	R	O	Q	A	G
				G			.

LETTER TRACKER

SECRET	A	B	C	D	E	F	G	H	I	J	K	L	M
REAL		K			—	—				B		—	

SECRET	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
REAL	—		—	G		—	—				—	—	—

A line in a box means that secret letter is not in the message.

What is the second word of the first sentence?

ANSWER _____ CHECK CODE (PAGE 12) _____

NONOGRAM | LEVEL ■■□ ADVANCED TYPICAL TIME: 12 MIN

THE SCRAMBLED SCREEN

An inventor hid a sketch inside this coded grid. Decode it to see what inspired her next project.

RULES Each number is a run of filled squares in that row or column, in order, with at least one empty square between runs. A 0 means the line stays empty. Fill in squares to reveal a picture, and put a dot in each square you know is empty.

	3	3	2	1 2	1 2	4 1	1 11	4 1	1 2	1 2	2	3	3
3													
1 1													
3													
7													
1													
1													
1													
1													
2 1 2													
2 1 2													
3 1 3													
3 1 3													
7													

What does the picture show? Write it in one word (6 letters).

ANSWER CHECK CODE (PAGE 12)

SYMBOL BALANCE | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

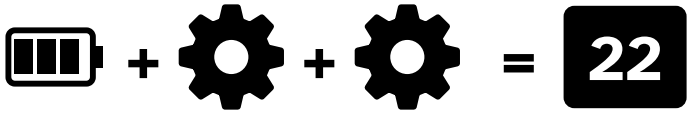
THE SPARK CHECK

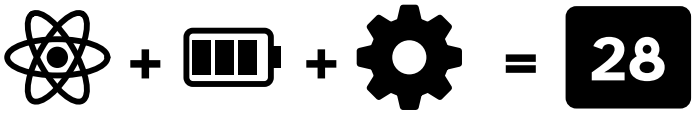
The control screen of the robot dog shows symbols instead of numbers after an update. Translate it back.

RULES | Each symbol stands for a whole number from 1 to 20. The same symbol always has the same value on every line. Every line balances: the left side equals the number on the right.

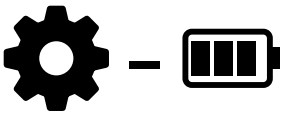
SYMBOL KEY

atom = ☐ battery = ☐ gear = ☐

1.  = 22

2.  = 28

3.  = 47

FIND () ×  = ☐

WORK SPACE

What is (gear – battery) × atom?

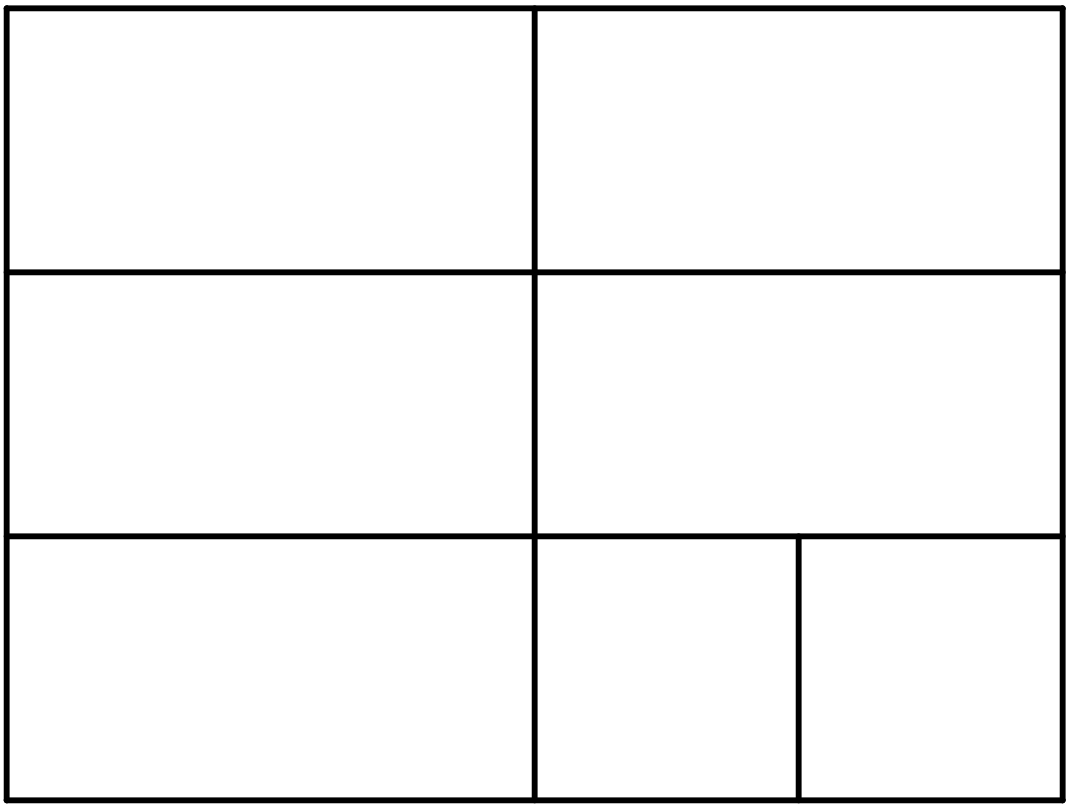
ANSWER ☐ CHECK CODE (PAGE 12) ☐

SHAPE COUNT | LEVEL ■■□ ADVANCED TYPICAL TIME: 9 MIN

CROSSED PATTERN

A rival agent counted the rectangles in the telescope logo and got it wrong. Do better.

RULES | Count every shape of the kind asked, big or small, sitting flat or standing on a corner. Shapes can overlap and can be made by joining pieces. A square counts as a rectangle, but a rectangle does not count as a square.



TALLY BY SIZE ☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

One box per size. More sizes than boxes? Put two sizes in one box.

How many rectangles are in the figure, squares included? Write the number.

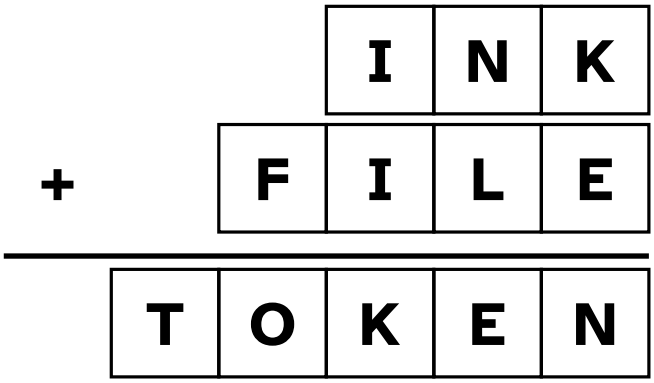
ANSWER ☐ CHECK CODE (PAGE 12) ☐

WORD MATH | LEVEL ■■□ ADVANCED TYPICAL TIME: 12 MIN

LOCKED SUM: TOKEN

Agent training test: this addition uses letters instead of digits, and it must add up. Find TOKEN.

RULES Each letter stands for one digit from 0 to 9, and different letters stand for different digits. No number starts with 0. Add as usual, from right to left: when a column adds up to 10 or more, carry into the next column.



CODE KEY

E	F	I	K	L	N	O	T
	9				5		

DIGITS USED (CROSS THEM OUT)

0	1	2	3	4	5	6	7	8	9

WORK SPACE

What number does TOKEN stand for?

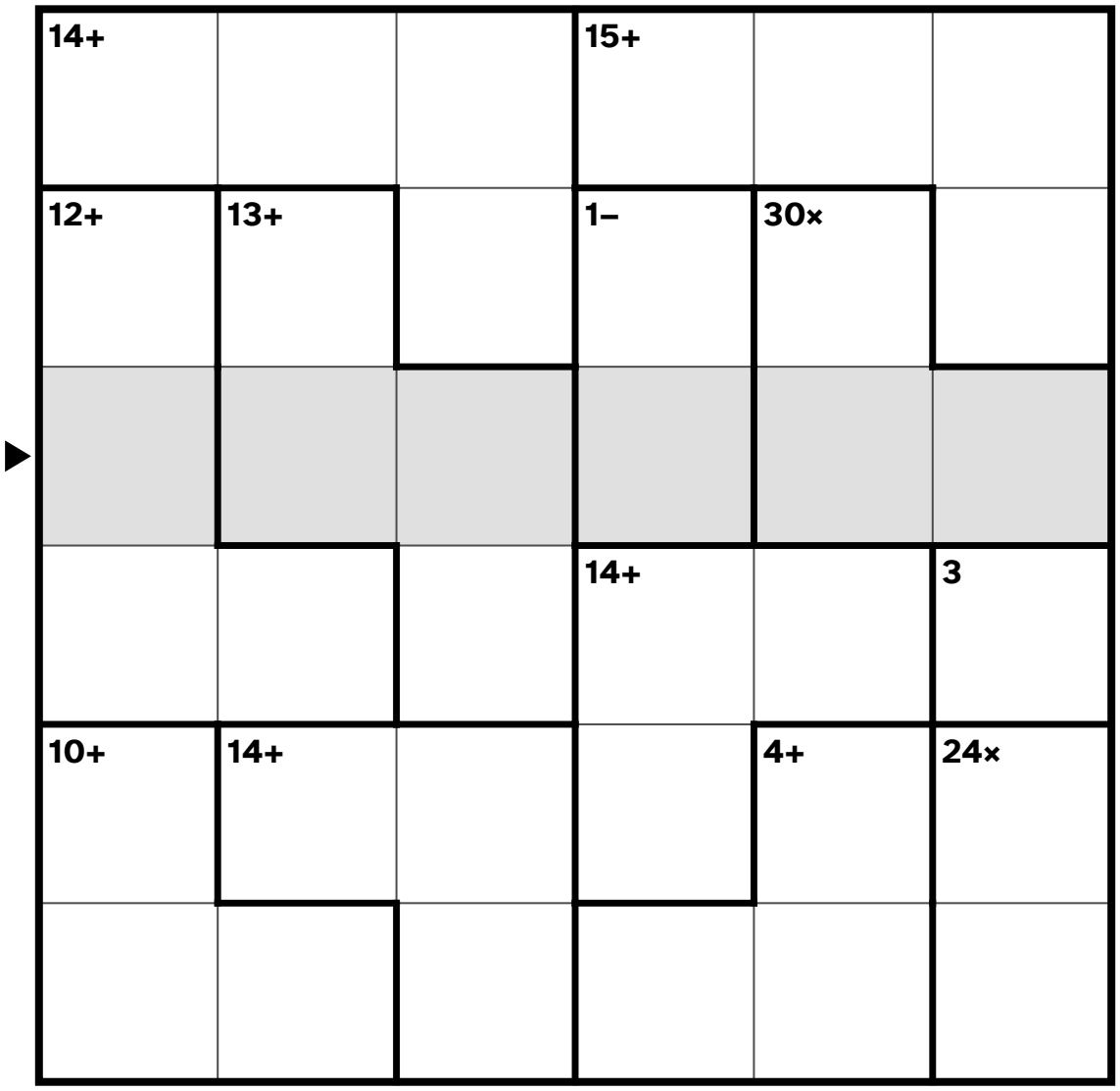
ANSWER CHECK CODE (PAGE 12)

CALC CAGES | LEVEL ■■■ ELITE TYPICAL TIME: 20 MIN

THE COBALT PANEL

A training safe in the Field Map office waits for you. Solve the grid to open it.

RULES Each row and each column uses every number from 1 to the grid size exactly once. A thick outline is a cage: its numbers must make the target with the sign shown. For a minus or divide cage, start with the bigger number. A number can repeat inside a cage if the two are not in the same row or column. A one-square cage with no sign gives you its number.



Read the shaded row from left to right. What number do you get?

ANSWER _____ CHECK CODE (PAGE 12) _____

CHECK YOUR ANSWER

Turn your answer into a check code. Then look at the two codes next to your mission number below.

If your code is one of them, your answer is almost certainly right. If it is not, check your work.

1. Write each letter as its number: A = 1, B = 2, C = 3 … Z = 26. A digit stays as it is.
2. Multiply the first by 1, the second by 2, the third by 3, and so on.
3. Add up the results. Then add ten times the mission number.

EXAMPLE · ANSWER "GEAR" IN MISSION 4

G	E	A	R
7 x 1 = 7	5 x 2 = 10	1 x 3 = 3	18 x 4 = 72

7 + 10 + 3 + 72 = 92

92 + 40 (mission 4 x 10) = 132

Check code: 132

CODES FOR MISSIONS 81 TO 90

One code in each pair is a decoy.

MISSION 81	931	919	MISSION 86	1113	1098
MISSION 82	930	957	MISSION 87	887	858
MISSION 83	881	853	MISSION 88	874	884
MISSION 84	951	988	MISSION 89	966	933
MISSION 85	1051	1029	MISSION 90	950	965

Code not next to your mission? Check your arithmetic first. If it is still missing, go back to the mission: one step went wrong somewhere. Take a hint if you want one.

HINT A

Read only the hint for your mission. Cover the rest with a sheet of paper.

DEPT 9 · BONUS FILES

M81 Mark every direct clue first. Then look hard at clue 2: it is the key, and the Either-Or technique unlocks it.

M82 Write WALRUS in the secret row, skipping any letter you already wrote, then add the rest of A to Z.

M83 Start with the clue that says nothing is correct, and cross out its digits everywhere.

M84 Start at the top: change one letter of BEAM so it matches the printed letters on rung 1.

M85 Write the given letters in every matching box first. Then look at DQDOR: think about its length, its known letters and its repeated letters.

M86 Start with row 4 and row 13: their clues let you mark squares before anything else is marked. A clue of 0, or a long run in a short line, leaves little room to slide.

M87 Find two lines where one holds everything the other has, plus a little more.

M88 Sort the rectangles by how many small squares wide and tall they are.

M89 Start with the thousands column, and list every way it can add up.

M90 Start with the one-square cages, then look for cages with very few number sets.

HINT B

Try Hint A first. Hint B does the first step for you, never the whole mission.

DEPT 9 · BONUS FILES

M81 **[Direct Clue]** Clue 1 gives a match: Sadie works with the robot Spark. Mark the O and **[Fill the Line]**.

M82 The secret alphabet begins W A L R U S B C D … Finish it, then find the first secret letter, V, in the secret row: it sits under the plain letter W.

M83 Clue 4 (092): 0, 2 and 9 are out.

M84 First step done: rung 1 is BEAT.

M85 The secret word DQDOR is AGAIN, so D is A, O is I and R is N.

M86 Row 4 (clue 7) has nothing marked yet. Slide the run both ways, then fill square 7.

M87 Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.

M88 There are 2 rectangles that are 1 by 1.

M89 **[Column Check]** Thousands column: 9 + the carry must end in O. It adds up only in these ways: O = 9, with no carry in and no carry out, but 9 is F; or O = 0, with a carry of 1 in and a carry of 1 out. So O = 0, the hundreds column carries 1 and the thousands column carries 1.

M90 The one-square cage at row 4, column 6 gives you a 3. Write it in, then cross that number off the rest of its row and its column.

Each solution names the technique it uses in bold. Learn the names: you will meet the same techniques again in harder missions.

MISSION 81 SECTOR ATLAS SORT-OUT

ANSWER: Kiara

Agent	Robot	Code Name	Floor
Isaac	Gear	Comet	Floor 3
Maya	Bolt	Otter	Floor 2
Sadie	Spark	Orca	Floor 1
Kiara	Rivet	Lynx	Floor 4

1. **[Direct Clue]** Clue 1 gives a match: Sadie works with the robot Spark. Mark the O and **[Fill the Line]**.
2. **[Direct Clue]** Clue 5 rules out one box: Isaac is not code-named Lynx. Put an X there.
3. **[Direct Clue]** Clue 6 gives a match: the agent with the robot Bolt works on Floor 2. Mark the O and **[Fill the Line]**.
4. **[Direct Clue]** Clue 7 gives a match: the agent code-named Orca works on Floor 1. Mark the O and **[Fill the Line]**.
5. **[Direct Clue]** Clue 8 gives a match: the agent code-named Otter works on Floor 2. Mark the O and **[Fill the Line]**.
6. **[Cross-Grid Transfer]** Sadie works with the robot Spark, so what is true of one is true of the other: the agent with the robot Spark does not work on Floor 2, so Sadie does not work on Floor 2.
7. **[Cross-Grid Transfer]** The agent with the robot Bolt works on Floor 2, so what is true of one is true of the other: the agent code-named Comet does not work on Floor 2, so the agent with the robot Bolt is not code-named Comet; the agent code-named Otter works on Floor 2, so the agent with the robot Bolt is code-named Otter. Mark the O and **[Fill the Line]**.

MISSION 81 (CONTINUED)

8. **[Cross-Grid Transfer]** Sadie works with the robot Spark, so what is true of one is true of the other: the agent with the robot Spark is not code-named Otter, so Sadie is not code-named Otter.
9. **[Either-Or]** Clue 2 says the agent on Floor 1 is either Sadie or Maya. So every other agent gets an X: Isaac does not work on Floor 1; Kiara does not work on Floor 1.
10. **[Cross-Grid Transfer]** The agent code-named Orca works on Floor 1, so what is true of one is true of the other: Isaac does not work on Floor 1, so Isaac is not code-named Orca; Kiara does not work on Floor 1, so Kiara is not code-named Orca.
11. **[Either-Or]** Clue 4 says the agent with the robot Gear is either Sadie or Isaac. So every other agent gets an X: Maya does not work with the robot Gear; Kiara does not work with the robot Gear.
12. **[Last Box]** Only one agent is left for the robot Gear, so Isaac works with the robot Gear. Mark the O and **[Fill the Line]**.
13. **[Cross-Grid Transfer]** The agent with the robot Bolt works on Floor 2, so what is true of one is true of the other: Isaac does not work with the robot Bolt, so Isaac does not work on Floor 2.
14. **[Cross-Grid Transfer]** The agent code-named Otter works on Floor 2, so what is true of one is true of the other: Isaac does not work on Floor 2, so Isaac is not code-named Otter.
15. **[Last Box]** Only one code name is left for Isaac, so Isaac is code-named Comet. Mark the O and **[Fill the Line]**.

MISSION 81 (CONTINUED)

16. **[Cross-Grid Transfer]** Sadie works with the robot Spark, so what is true of one is true of the other: Sadie is not code-named Comet, so the agent with the robot Spark is not code-named Comet.

17. **[Cross-Grid Transfer]** Isaac works with the robot Gear, so what is true of one is true of the other: Isaac is code-named Comet, so the agent with the robot Gear is code-named Comet. Mark the O and **[Fill the Line]**.

18. **[Cross-Grid Transfer]** The agent code-named Orca works on Floor 1, so what is true of one is true of the other: the agent with the robot Gear is not code-named Orca, so the agent with the robot Gear does not work on Floor 1.

19. **[Order Bounds]** Clue 3 sets an order, so the agent with the robot Gear does not work on Floor 4; Kiara does not work on Floor 2, because the agent with the robot Gear can be no lower than Floor 3; Kiara does not work on Floor 3, because the agent with the robot Gear can be no lower than Floor 3.

20. **[Last Box]** Only one floor is left for Kiara, so Kiara works on Floor 4. Mark the O and **[Fill the Line]**.

21. **[Last Box]** Only one floor is left for Isaac, so Isaac works on Floor 3. Mark the O and **[Fill the Line]**.

22. **[Last Box]** Only one floor is left for Sadie, so Sadie works on Floor 1. Mark the O and **[Fill the Line]**.

23. **[Last Box]** Only one floor is left for Maya, so Maya works on Floor 2. Mark the O and **[Fill the Line]**.

24. **[Last Box]** Only one floor is left for the agent with the robot Gear, so the agent with the robot Gear works on Floor 3. Mark the O and **[Fill the Line]**.

MISSION 81 (CONTINUED)

25. **[Cross-Grid Transfer]** Sadie works with the robot Spark, so what is true of one is true of the other: Sadie works on Floor 1, so the agent with the robot Spark works on Floor 1. Mark the O and **[Fill the Line]**.

26. **[Last Box]** Only one floor is left for the agent with the robot Rivet, so the agent with the robot Rivet works on Floor 4. Mark the O and **[Fill the Line]**.

27. **[Cross-Grid Transfer]** The agent with the robot Bolt works on Floor 2, so what is true of one is true of the other: Maya works on Floor 2, so Maya works with the robot Bolt. Mark the O and **[Fill the Line]**.

28. **[Last Box]** Only one robot is left for Kiara, so Kiara works with the robot Rivet. Mark the O and **[Fill the Line]**.

29. **[Cross-Grid Transfer]** The agent code-named Orca works on Floor 1, so what is true of one is true of the other: Maya does not work on Floor 1, so Maya is not code-named Orca.

30. **[Last Box]** Only one agent is left for the code name Orca, so Sadie is code-named Orca. Mark the O and **[Fill the Line]**.

31. **[Cross-Grid Transfer]** Sadie works with the robot Spark, so what is true of one is true of the other: Sadie is not code-named Lynx, so the agent with the robot Spark is not code-named Lynx.

32. **[Last Box]** Only one code name is left for the agent with the robot Spark, so the agent with the robot Spark is code-named Orca. Mark the O and **[Fill the Line]**.

33. **[Last Box]** Only one code name is left for the agent with the robot Rivet, so the agent with the robot Rivet is code-named Lynx. Mark the O and **[Fill the Line]**.

MISSION 81 (CONTINUED)

34. **[Cross-Grid Transfer]** The agent code-named Otter works on Floor 2, so what is true of one is true of the other: Maya works on Floor 2, so Maya is code-named Otter. Mark the O and **[Fill the Line]**.

35. **[Last Box]** Only one code name is left for Kiara, so Kiara is code-named Lynx. Mark the O and **[Fill the Line]**.

36. **[Cross-Grid Transfer]** Isaac is code-named Comet, so what is true of one is true of the other: Isaac works on Floor 3, so the agent code-named Comet works on Floor 3. Mark the O and **[Fill the Line]**.

37. **[Last Box]** Only one floor is left for the agent code-named Lynx, so the agent code-named Lynx works on Floor 4. Mark the O and **[Fill the Line]**.

38. The grid is now full: Kiara is code-named Lynx. The answer is Kiara.

MISSION 82 CODE NAME: BROKEN BEACON

ANSWER: WATCH

V	W	P	L	C
W	**A**	**T**	**C**	**H**
W	A	J	T	U
A	B	O	V	E
U	D	B	C	P
E	I	G	H	T

P	C	U
T	H	E
L	W	A	D	I
C	A	B	I	N

O	F	Y
S	K	Y

Secret alphabet: WALRUSBCDEFGHIJKMNOPQTVXYZ. Answer: WATCH.

1. **[Build the Key Alphabet]** WALRUS without repeats is WALRUS; the unused letters follow, so the secret alphabet starts WALRUSBCDE.

2. **[Reverse Lookup]** Find each secret letter in the secret row and read the plain letter above it: V sits under W.

MISSION 82 (CONTINUED)

3. **[Word Check]** The message reads: WATCH THE SKY ABOVE CABIN EIGHT. Every word is a real word, so the first word is WATCH.

MISSION 83 THE RUSTY TUMBLER

ANSWER: 615

1	**7**	**2**	**5**	1 right spot
2	**1**	**4**	**7**	1 wrong spot
3	**9**	**6**	**5**	1 right spot, 1 wrong spot
4	**0**	**9**	**2**	nothing correct

CODE **6** **1** **5**

1. **[Nothing Is Correct]** Clue 4 (092): 0, 2 and 9 are out. **[Count the Digits]** Clue 3 (965): 5 and 6 are in the code.

2. **[Count the Digits]** Clue 1 (725): 7 is out. **[Right Spot]** Clue 1 (725): 5 goes in the third spot.

3. **[Wrong Spot]** Clue 2 (147): 1 is not in the first spot; 4 is not in the second spot.

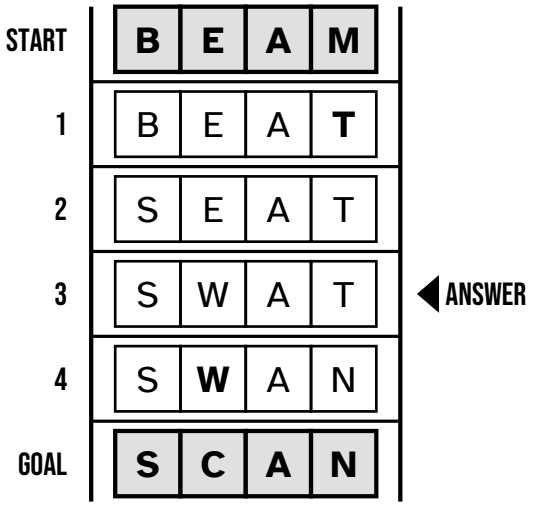
4. **[Place Count]** Clue 3 (965): 6 goes in the first spot. **[Only Spot Left]** 4 has no spot left, so it is out.

5. **[Count the Digits]** Clue 2 (147): 1 goes in the second spot.

6. So the code is 615.

MISSION 84 SIGNAL PATH: BEAM TO SCAN

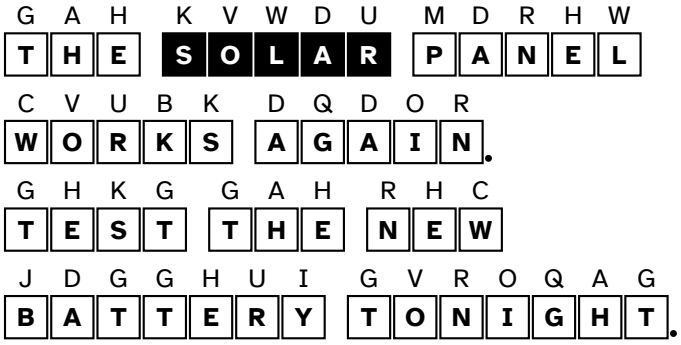
ANSWER: SWAT



1. **[Step Down]** Change one letter of BEAM: only BEAT fits the printed letters on rung 1.
2. **[Step Up]** Work up from SCAN: only SWAN fits the printed letters on rung 4.
3. **[Count the Gap]** Rung 3 must be one letter from SWAN and still reach BEAT in the rungs left between them. Of the words one letter from SWAN, only SWAT is few enough changes from BEAT, so rung 3 is SWAT.
4. **[Squeeze Play]** Rung 2 must be one letter from BEAT and from SWAT, so it is SEAT.
5. The word on rung 3 is SWAT.

MISSION 85 RADIO LOG: COBALT TIDE

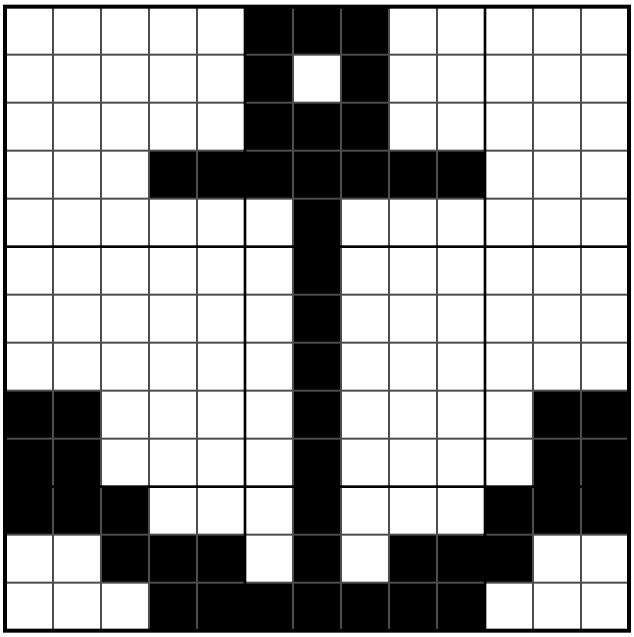
ANSWER: SOLAR



1. **[Given Letters]** Start with the letters you are given: B is K, J is B and Q is G. Write each one under every matching secret letter, and in the tracker.
2. **[Word Shape]** DQDOR now reads ?G???. Its length, its known letters and its pattern fit only one everyday word, AGAIN. So D is A, O is I and R is N.
3. **[Word Shape]** GVROQAG now reads ??NIG??. Its length, its known letters and its pattern fit only one everyday word, TONIGHT. So G is T, V is O and A is H.
4. **[Word Shape]** RHC now reads N??. Its length, its known letters and its pattern fit only one everyday word, NEW (NOT and NOW would need O, which is already taken). So H is E and C is W.
5. Keep going with **[Word Shape]** and **[Fill the Gaps]**, one word at a time: CVUBK, which reads WO?K?, is WORKS, KVWDU, which reads SO?AR, is SOLAR and MDRHW, which reads ?ANEL, is PANEL, and the other words fill in the same way.
6. **[Read It Back]** The message reads: THE SOLAR PANEL WORKS AGAIN. TEST THE NEW BATTERY TONIGHT. Every word is real, so the second word of the first sentence is SOLAR.

MISSION 86 THE SCRAMBLED SCREEN

ANSWER: ANCHOR



1. Count squares from the left in a row and from the top in a column. **[Overlap]** Row 4 (clue 7) has nothing marked yet. Slide the run both ways, then fill square 7.
2. **[Overlap]** Row 13 (clue 7) has nothing marked yet. Slide the run both ways, then fill square 7.
3. **[Slide Test]** Column 7 (clue 1 11): push both runs to one end, then to the other, around the marks. Fill squares 1, 3 and 5-12 and put a dot in square 2.
4. **[One Run]** Row 1 (clue 3) has one run. It must cover its filled squares and miss the dots. Put a dot in squares 1-4 and 10-13.
5. **[One Run]** Row 3 (clue 3) has one run. It must cover its filled squares and miss the dots. Put a dot in squares 1-4 and 10-13.
6. **[Line Done]** Row 5 (clue 1) already has all its filled squares, so put a dot in squares 1-6 and 8-13.
7. **[Line Done]** Row 6 (clue 1) already has all its filled squares, so put a dot in squares 1-6 and 8-13.

MISSION 86 (CONTINUED)

8. **[Line Done]** Row 7 (clue 1) already has all its filled squares, so put a dot in squares 1-6 and 8-13.
9. **[Line Done]** Row 8 (clue 1) already has all its filled squares, so put a dot in squares 1-6 and 8-13.
10. **[One Run]** Column 1 (clue 3) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Fill square 11 and put a dot in squares 2 and 4.
11. **[One Run]** Column 2 (clue 3) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Fill square 11 and put a dot in squares 2 and 4.
12. **[One Run]** Column 3 (clue 2) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Put a dot in squares 2 and 4.
13. **[Slide Test]** Column 6 (clue 4 1): push both runs to one end, then to the other, around the marks. Fill squares 1-4.
14. **[Slide Test]** Column 8 (clue 4 1): push both runs to one end, then to the other, around the marks. Fill squares 1-4.
15. **[One Run]** Column 11 (clue 2) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Put a dot in squares 2 and 4.
16. **[One Run]** Column 12 (clue 3) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Fill square 11 and put a dot in squares 2 and 4.

MISSION 86 (CONTINUED)

17. **[One Run]** Column 13 (clue 3) has one run and nothing filled yet. The run can only sit where the gap between the dots is long enough for it: test every place it fits. Fill square 11 and put a dot in squares 2 and 4.

18. **[Line Done]** Row 1 (clue 3) already has all its filled squares, so put a dot in squares 5 and 9.

19. **[Line Done]** Row 2 (clue 1 1) already has all its filled squares, so put a dot in squares 4-5 and 9-10.

20. **[Line Done]** Row 3 (clue 3) already has all its filled squares, so put a dot in squares 5 and 9.

21. **[One Run]** Row 4 (clue 7) has one run. It must cover its filled squares and miss the dots. Fill squares 4-5 and 9-10.

22. **[Match the Runs]** Row 11 (clue 3 1 3): match each filled square to its run, then slide each run. Fill squares 3 and 11 and put a dot in squares 4-6 and 8-10.

23. **[One Run]** Column 3 (clue 2) has one run. It must cover its filled squares and miss the dots. Put a dot in squares 9 and 13.

24. **[One Run]** Column 11 (clue 2) has one run. It must cover its filled squares and miss the dots. Put a dot in squares 9 and 13.

25. **[One Run]** Row 13 (clue 7) has one run. It must cover its filled squares and miss the dots. Fill squares 4-6 and 8-10 and put a dot in squares 1-2 and 12-13.

26. **[One Run]** Column 1 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 10.

27. **[One Run]** Column 2 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 10.

MISSION 86 (CONTINUED)

28. **[Slide Test]** Column 4 (clue 1 2): push both runs to one end, then to the other, around the marks. Fill square 12 and put a dot in squares 9-10.

29. **[Slide Test]** Column 5 (clue 1 2): push both runs to one end, then to the other, around the marks. Fill square 12 and put a dot in squares 9-10.

30. **[Line Done]** Column 6 (clue 4 1) already has all its filled squares, so put a dot in squares 9-10 and 12.

31. **[Line Done]** Column 8 (clue 4 1) already has all its filled squares, so put a dot in squares 9-10 and 12.

32. **[Slide Test]** Column 9 (clue 1 2): push both runs to one end, then to the other, around the marks. Fill square 12 and put a dot in squares 9-10.

33. **[Slide Test]** Column 10 (clue 1 2): push both runs to one end, then to the other, around the marks. Fill square 12 and put a dot in squares 9-10.

34. **[One Run]** Column 12 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 10.

35. **[One Run]** Column 13 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 10.

36. **[Line Done]** Row 10 (clue 2 1 2) already has all its filled squares, so put a dot in squares 3 and 11.

37. **[One Run]** Column 3 (clue 2) has one run. It must cover its filled squares and miss the dots. Fill square 12.

38. **[One Run]** Column 11 (clue 2) has one run. It must cover its filled squares and miss the dots. Fill square 12.

MISSION 86 (CONTINUED)

39. **[Line Done]** Row 12 (clue 3 1 3) already has all its filled squares, so put a dot in squares 1-2 and 12-13.

40. **[One Run]** Column 1 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 9.

41. **[One Run]** Column 2 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 9.

42. **[One Run]** Column 12 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 9.

43. **[One Run]** Column 13 (clue 3) has one run. It must cover its filled squares and miss the dots. Fill square 9.

44. Every square is now marked. The finished picture shows an anchor.

MISSION 87 THE SPARK CHECK

ANSWER: 75

atom = 15 battery = 4 gear = 9

1 🔋 **+** ⚙ **+** ⚙ **= 22**

2 ⚛ **+** 🔋 **+** ⚙ **= 28**

3 ⚛ **+** ⚛ **+** 🔋 **+** 🔋 **+** ⚙ **= 47**

? (⚙ **–** 🔋) **×** ⚛ **= 75**

1. **[Compare Scales]** Line 3 holds everything in line 2 plus an atom and a battery, so atom + battery = 47 – 28 = 19.

2. **[Compare Scales]** Line 2 is atom + battery + gear = 28, and we know that atom + battery = 19, so gear = 28 – 19 = 9.

3. **[Swap In]** Put gear = 9 into line 1: battery + 9 + 9 = 22, so the battery is 4.

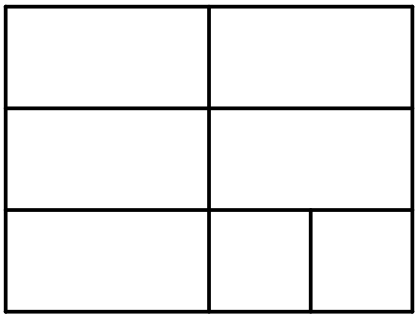
MISSION 87 (CONTINUED)

4. **[Swap In]** Put battery = 4 and gear = 9 into line 2: atom + 4 + 9 = 28, so the atom is 15.

5. So (gear – battery) × atom = (9 – 4) × 15 = 75.

MISSION 88 CROSSED PATTERN

ANSWER: 21



Answer: 21 rectangles. Sort the rectangles by size, counting in small squares: 1 by 1: 2; 1 by 2: 6; 1 by 3: 1; 1 by 4: 3. And the rest: 2 by 2: 4; 2 by 3: 2; 2 by 4: 2; 3 by 4: 1.

1. **[Width x Height]** Sort the rectangles by size, counting in small squares: 1 by 1: 2; 1 by 2: 6; 1 by 3: 1; 1 by 4: 3.

2. **[Width x Height]** And the rest: 2 by 2: 4; 2 by 3: 2; 2 by 4: 2; 3 by 4: 1.

3. Add them all: 2 + 6 + 1 + 3 + 4 + 2 + 2 + 1 = 21 rectangles.

MISSION 89 LOCKED SUM: TOKEN

ANSWER: 10325

		I 6	N 5	K 3
+	F 9	I 6	L 7	E 2
T 1	O 0	K 3	E 2	N 5

E=2, F=9*, I=6, K=3, L=7, N=5*, O=0, T=1
* given

1. Write the given digits under their letters: F = 9, N = 5.
2. **[Column Check]** Thousands column: 9 + the carry must end in O. It adds up only in these ways: O = 9, with no carry in and no carry out, but 9 is F; or O = 0, with a carry of 1 in and a carry of 1 out. So O = 0, the hundreds column carries 1 and the thousands column carries 1.
3. **[Column Check]** T, the first digit of TOKEN (in the ten-thousands column), is the carry from the thousands column. So T = 1.
4. **[Suppose and Crash]** Digits still possible for E: 2, 3, 4, 6, 7, 8. If E = 2, it fits. If E = 3, the ones column gives K = 2 (2 + 3 = 5), not carrying, then the tens column gives L = 8 (5 + 8 = 13), carrying 1, then the hundreds column must make 12, so I + I would have to make 11, and no digit does that. If E = 4, the ones column gives K = 1 (1 + 4 = 5), but 1 is T. If E = 6, the ones column gives K = 9 (9 + 6 = 15), but 9 is F. If E = 7, the ones column gives K = 8 (8 + 7 = 15), carrying 1, then the tens column gives L = 1 (5 + 1 + 1 (carried) = 7), but 1 is T. If E = 8, the ones column gives K = 7 (7 + 8 = 15), carrying 1, then the tens column gives L = 2 (5 + 2 + 1 (carried) = 8), not carrying, then the hundreds column must make 17, so I + I would have to make 17, and no digit does that. So E = 2.

MISSION 89 (CONTINUED)

5. **[Column Check]** Ones column: K + 2 must end in 5: 3 + 2 = 5. So K = 3 and the ones column does not carry. **[Column Check]** Tens column: 5 + L must end in 2: 5 + 7 = 12. So L = 7 and the tens column carries 1. **[Column Check]** Hundreds column: I + I + 1 (carried) must end in 3 and carry 1: 6 + 6 + 1 (carried) = 13. So I = 6.
6. So TOKEN = 10325.

MISSION 90 THE COBALT PANEL

ANSWER: 462351

14+ 6	1	4	15+ 5	3	2
12+ 1	13+ 4	3	1– 2	30× 6	5
4	6	2	3	5	1
5	2	1	14+ 6	4	3 3
10+ 2	14+ 3	5	4	4+ 1	24× 6
3	5	6	1	2	4

1. **[Free Number]** The one-square cage at row 4, column 6 gives you a 3. Write it in, then cross that number off the rest of its row and its column.
2. **[Cage Options]** The 14+ cage at row 5, column 2: with the numbers its squares can still hold, only {2, 6, 6}, {3, 5, 6}, {4, 4, 6} or {4, 5, 5} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 5, column 2 and row 6, column 3 can only be 3, 4, 5 or 6; row 5, column 3 can only be 2, 3, 4, 5 or 6.

MISSION 90 (CONTINUED)

3. **[Cage Options]** The 30× cage at row 2, column 5: with the numbers its squares can still hold, only {1, 5, 6} or {2, 3, 5} work. So row 2, column 5 and row 3, column 5 can only be 1, 2, 3, 5 or 6; row 3, column 6 can only be 1, 2, 5 or 6.
4. **[Cage Options]** The 4+ cage at row 5, column 5: with the numbers its squares can still hold, only {1, 1, 2} works. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 5, column 5 and row 6, column 4 can only be 1; row 6, column 5 can only be 2.
5. **[Last Option]** Only 1 is left for row 5, column 5. Write it in and cross it off its row and column.
6. **[Last Option]** Only 1 is left for row 6, column 4. Write it in and cross it off its row and column.
7. **[Last Option]** Only 2 is left for row 6, column 5. Write it in and cross it off its row and column.
8. **[Cage Options]** The 10+ cage at row 5, column 1: with the numbers its squares can still hold, only {2, 3, 5} or {3, 3, 4} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 5, column 1 can only be 2 or 3; row 6, column 1 can only be 3, 4 or 5; row 6, column 2 can only be 3 or 5.
9. **[Cage Options]** The 30× cage at row 2, column 5: with the numbers its squares can still hold, only {1, 5, 6} or {2, 3, 5} work. So row 3, column 6 can only be 1 or 2.

MISSION 90 (CONTINUED)

10. **[Cage Options]** The 14+ cage at row 4, column 4: with the numbers its squares can still hold, only {2, 6, 6}, {3, 5, 6}, {4, 4, 6} or {4, 5, 5} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 5, column 4 can only be 3, 4, 5 or 6.
11. **[Cage Options]** The 24× cage at row 5, column 6: with the numbers its squares can still hold, only {4, 6} works. So row 5, column 6 and row 6, column 6 can only be 4 or 6.
12. **[Cage Claim]** Every way to fill the 10+ cage at row 5, column 1 puts a 3 in row 6, so no other square in row 6 can hold a 3. So row 6, column 3 can only be 4, 5 or 6.
13. **[Cage Claim]** Every way to fill the 30× cage at row 2, column 5 puts a 5 in column 5, so no other square in column 5 can hold a 5. So row 1, column 5 can only be 3, 4 or 6; row 4, column 5 can only be 4 or 6.
14. **[Cage Options]** The 14+ cage at row 4, column 4: with the numbers its squares can still hold, only {2, 6, 6}, {3, 5, 6} or {4, 4, 6} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 4, column 4 can only be 2, 5 or 6; row 5, column 4 can only be 3, 4 or 6.
15. **[Cage Claim]** Every way to fill the 14+ cage at row 4, column 4 puts a 6 in row 4, so no other square in row 4 can hold a 6. So row 4, column 1; row 4, column 2 and row 4, column 3 can only be 1, 2, 4 or 5.
16. **[Cage Claim]** Every way to fill the 24× cage at row 5, column 6 puts a 4 in column 6, so no other square in column 6 can hold a 4. So row 1, column 6 and row 2, column 6 can only be 1, 2, 5 or 6.

MISSION 90 (CONTINUED)

17. **[Cage Claim]** Every way to fill the 24× cage at row 5, column 6 puts a 6 in column 6, so no other square in column 6 can hold a 6. So row 1, column 6 and row 2, column 6 can only be 1, 2 or 5.

18. **[Cage Options]** The 15+ cage at row 1, column 4: with the numbers its squares can still hold, only {1, 3, 5, 6}, {1, 4, 5, 5}, {2, 2, 5, 6} or {2, 3, 5, 5} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 1, column 4 can only be 2, 3, 5 or 6.

19. **[Test and Reject]** Test the 14+ cage at row 5, column 2: if row 5, column 2 were 4, every way to finish the cage would leave row 5, column 6 with no number left; if row 5, column 2 were 6, every way to finish the cage would leave row 5 with no place for a 5; if row 5, column 3 were 2, every way to finish the cage would leave row 5 with no place for a 5; if row 5, column 3 were 6, every way to finish the cage would leave row 5 with no place for a 5 or row 5, column 6 with no number left; if row 6, column 3 were 4, every way to finish the cage would leave row 5, column 6 with no number left. So row 5, column 2 can only be 3 or 5; row 5, column 3 can only be 3, 4 or 5; row 6, column 3 can only be 5 or 6.

20. **[Only Spot]** In row 5, a 2 can only go at row 5, column 1. Write it in and cross it off its row and column.

21. **[Cage Options]** The 10+ cage at row 5, column 1: with the numbers its squares can still hold, only {2, 3, 5} works. So row 6, column 1 can only be 3 or 5.

22. **[Only Spot]** In row 6, a 4 can only go at row 6, column 6. Write it in and cross it off its row and column.

23. **[Last Option]** Only 6 is left for row 5, column 6. Write it in and cross it off its row and column.

MISSION 90 (CONTINUED)

24. **[Only Spot]** In row 6, a 6 can only go at row 6, column 3. Write it in and cross it off its row and column.

25. **[Cage Options]** The 14+ cage at row 5, column 2: with the numbers its squares can still hold, only {3, 5, 6} works. So row 5, column 3 can only be 3 or 5.

26. **[Only Spot]** In row 5, a 4 can only go at row 5, column 4. Write it in and cross it off its row and column.

27. **[Cage Options]** The 12+ cage at row 2, column 1: with the numbers its squares can still hold, only {1, 1, 4, 6}, {1, 2, 3, 6}, {1, 2, 4, 5} or {1, 3, 4, 4} work. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 4, column 2 can only be 1, 2 or 4.

28. **[Cage Options]** The 14+ cage at row 4, column 4: with the numbers its squares can still hold, only {4, 4, 6} works. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 4, column 4 can only be 6; row 4, column 5 can only be 4.

29. **[Last Option]** Only 6 is left for row 4, column 4. Write it in and cross it off its row and column.

30. **[Last Option]** Only 4 is left for row 4, column 5. Write it in and cross it off its row and column.

31. **[Cage Options]** The 1– cage at row 2, column 4: with the numbers its squares can still hold, only {2, 3} works. So row 2, column 4 and row 3, column 4 can only be 2 or 3.

32. **[Only Spot]** In column 4, a 5 can only go at row 1, column 4. Write it in and cross it off its row and column.

33. **[Only Spot]** In column 6, a 5 can only go at row 2, column 6. Write it in and cross it off its row and column.

MISSION 90 (CONTINUED)

34. **[Only Spot]** In column 5, a 5 can only go at row 3, column 5. Write it in and cross it off its row and column.

35. **[Cage Options]** The 15+ cage at row 1, column 4: with the numbers its squares can still hold, only {2, 3, 5, 5} works. Two equal numbers can never share a row or column, so each set fits in only some ways. So row 1, column 5 can only be 3; row 1, column 6 can only be 2.

36. **[Last Option]** Only 3 is left for row 1, column 5. Write it in and cross it off its row and column.

37. **[Last Option]** Only 2 is left for row 1, column 6. Write it in and cross it off its row and column.

38. **[Last Option]** Only 6 is left for row 2, column 5. Write it in and cross it off its row and column.

39. **[Last Option]** Only 1 is left for row 3, column 6. Write it in and cross it off its row and column.

40. **[Cage Options]** The 12+ cage at row 2, column 1: with the numbers its squares can still hold, only {1, 2, 3, 6} or {1, 2, 4, 5} work. So row 2, column 1 can only be 1 or 3; row 3, column 1 can only be 4 or 6; row 4, column 2 can only be 2.

41. **[Last Option]** Only 2 is left for row 4, column 2. Write it in and cross it off its row and column.

42. **[Cage Options]** The 14+ cage at row 1, column 1: with the numbers its squares can still hold, only {1, 3, 4, 6} works. So row 2, column 3 can only be 3.

43. **[Last Option]** Only 3 is left for row 2, column 3. Write it in and cross it off its row and column.

MISSION 90 (CONTINUED)

44. **[Last Option]** Only 1 is left for row 2, column 1. Write it in and cross it off its row and column.

45. **[Last Option]** Only 4 is left for row 2, column 2. Write it in and cross it off its row and column.

46. **[Last Option]** Only 2 is left for row 2, column 4. Write it in and cross it off its row and column.

47. **[Last Option]** Only 3 is left for row 3, column 4. Write it in and cross it off its row and column.

48. **[Last Option]** Only 6 is left for row 3, column 2. Write it in and cross it off its row and column.

49. **[Last Option]** Only 1 is left for row 1, column 2. Write it in and cross it off its row and column.

50. **[Last Option]** Only 4 is left for row 1, column 3. Write it in and cross it off its row and column.

51. **[Last Option]** Only 6 is left for row 1, column 1. Write it in and cross it off its row and column.

52. **[Last Option]** Only 4 is left for row 3, column 1. Write it in and cross it off its row and column.

53. **[Last Option]** Only 2 is left for row 3, column 3. Write it in and cross it off its row and column.

54. **[Last Option]** Only 5 is left for row 4, column 1. Write it in and cross it off its row and column.

55. **[Last Option]** Only 1 is left for row 4, column 3. Write it in and cross it off its row and column.

MISSION 90 (CONTINUED)

56. **[Last Option]** Only 5 is left for row 5, column 3. Write it in and cross it off its row and column.

57. **[Last Option]** Only 3 is left for row 5, column 2. Write it in and cross it off its row and column.

58. **[Last Option]** Only 3 is left for row 6, column 1. Write it in and cross it off its row and column.

59. **[Last Option]** Only 5 is left for row 6, column 2. Write it in and cross it off its row and column.

60. Every square is now filled. The shaded row reads 462351.